

G1-Hard-3B

69.36*

70.64*

48.50

17.43

53.30

mup. We study the role of SFT as a cold-start mechanism for RL, evaluating its impact on both performance and response behavior. To isolate the effect of SFT, we compare two variants: G1-Zero-3B trained directly from the base model Qwen2.5-72B-Instruct with RL, and G1-3B that initializes RL with the CoT-SFT checkpoint. As shown in Figure 1, training RL directly from the base model achieves consistently strong performance, aligning with results in Deepseek-D1-Zero [12]. Meanwhile,

